Ankitha Betala

Aerospace Structural Analysis Engineer
FEA — Thermal Analysis — Fluid–Structure Interaction

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023–2025	M.Tech (Aero Thermal)	Indian Institute of Technology, Kanpur	6.5/10
2023	B.E. (Aeronautical Engineering)	Hindusthan Institute of Technology, Coimbatore	8.86/10
2019	Board of Intermediate Education(XII)	Sri Chaitanya Jr. Mahila Kalasala, Vijayawada	9.92/10

Technical Skills

- **FEA:** ANSYS Mechanical, Linear Static Analysis, Nonlinear Structural Analysis, Contact Mechanics, Stress and Deformation Analysis, Structural Integrity Assessment
- **Thermal & Multiphysics:** Thermal Analysis, Thermal–Structural Analysis, Conjugate Heat Transfer (CHT), Fluid–Structure Interaction (FSI)
- **Engineering Methods:** Finite Element Method (FEM), Mesh Convergence, Material Modelling, Boundary Conditions, Contact Formulation
- **Programming & Data Analysis:** MATLAB, Python, NI LabVIEW
- **HPC & Computing:** High Performance Computing (HPC), Linux Environment, Remote Cluster Computing, MobaXterm
- **Tools:** ANSYS Workbench

Professional Experience

Engineer II (*Jan'26–Present*)
INVENTGRID INDIA PRIVATE LIMITED (IG Defence)
Department: New Development – CFD Analysis

- **JWALA Defence Project**
  - Performed linear and nonlinear structural analyses of rocket motor components using ANSYS Mechanical to evaluate stress distribution, deformation and structural response under applied operating loads.
  - Defined material properties, boundary conditions, contact formulations and loading conditions to assess component behaviour under operating conditions.
  - Evaluated von Mises stress, total deformation and critical stress concentrations to identify potential structural failure locations.
  - Analysed threaded connections and contact interfaces to investigate relative sliding, contact separation and potential component detachment.
  - Conducted fluid–structure interaction (FSI) analysis of a rocket nozzle to evaluate pressure-induced structural deformation and stress response.
  - Interpreted simulation results and prepared engineering reports to support structural assessment and design reviews.
- **KAL RATO Booster Project**
  - Post-processed static test data from D1 and D3 motor tests to extract thrust-time and chamber-pressure histories and evaluate propulsion performance for a 250 kg-class RATO system.
  - Calculated required thrust and impulse for rail inclinations of 15°–20° and target speeds of 160–190 km/h.
  - Evaluated first-three-second impulse values of 14,571.429 N s (D1) and 14,339.812 N s (D3) against the nominal requirement of 14,040.189 N s.
  - Performed linear and nonlinear structural analyses of rocket motor components using ANSYS Mechanical to evaluate stress distribution and deformation.
  - Analysed threaded connections and contact interfaces to assess potential sliding, separation and component detachment.
  - Conducted structural analysis of the motor test rig, obtaining a maximum reported von Mises stress of 16.22 MPa and maximum total deformation of 0.042461 mm.
  - Performed thermal and FSI analyses to assess temperature distribution, pressure-induced deformation and structural response in propulsion components.

Research Interests

- Structural Integrity, Finite Element Methods (FEM) and Nonlinear Contact Mechanics
- Thermal–Structural Analysis and Thermomechanical Behaviour
- Fluid–Structure Interaction (FSI) and Multiphysics Modelling
- Structural Dynamics, Modal Analysis and Fatigue Behaviour

- Stress Analysis, Failure Assessment and Design Optimisation
- Aerospace Propulsion Components, Gas Turbines and Thermal Management

Key Projects

M.Tech Thesis: Conjugate Heat Transfer Analysis of a Circular Nozzle with Passive Cooling (*Aug'2024–2025*)
Proposed by: Aeronautical Development Establishment (ADE), DRDO
Supervisor: Dr. Abhijith Kushari — Indian Institute of Technology Kanpur

- Conducted numerical investigation of conjugate heat transfer in a circular nozzle with passive cooling using **ANSYS Fluent**.
- Investigated coupled fluid-flow and solid heat-conduction behaviour to evaluate nozzle thermal performance.
- Analysed temperature distribution, wall heat flux, cooling effectiveness and thermal gradients to identify thermally critical regions.
- Performed simulations on High Performance Computing (HPC) clusters using Linux-based remote computing environments.

Signal Processing and Modal Analysis Using MATLAB
Independent Computational Studies — Measurement and Data Analysis

- Developed MATLAB workflows using auto-correlation, cross-correlation and FFT to investigate periodic behaviour, dominant frequency components and time-delay relationships in synthetic pressure signals.
- Implemented Proper Orthogonal Decomposition (POD) and Spectral Proper Orthogonal Decomposition (SPOD) to identify energetic spatial modes and frequency-dependent coherent structures in synthetic two-dimensional velocity fields.
- Evaluated modal energy distribution, cumulative energy and SPOD spectra; the generated examples showed a leading POD mode with approximately 67% energy and a dominant SPOD spectral peak near 78.12 Hz.

Welding of Dissimilar Aluminium Alloys (Al7075–Al6063) (*Jan'23–May'23*)
Queen India Engineering Services, Chennai

- Investigated TIG welding of dissimilar aluminium alloys using ER4043 filler material.
- Evaluated mechanical properties through hardness, impact and microstructural analysis.
- Performed non-destructive testing (NDT) to identify surface and internal defects.

Course Projects

- **Temperature Measurement of Wind-Driven Pool Fire Using LabVIEW** (*Apr'2024*)
  Guide: Dr. Kamal Poddar — Department of Aerospace Engineering, IIT Kanpur
  - Developed a LabVIEW interface for temperature acquisition and monitoring in pool-fire experiments.
  - Performed data acquisition, processing and temperature measurement analysis.
- **Transient Thermal Analysis of a Sphere with an Enclosure** (*Jan'2024–May'2024*)
  Guide: Dr. Mohammad Ibrahim Sugarno — Department of Aerospace Engineering, IIT Kanpur
  - Conducted transient thermal analysis using ANSYS Workbench.
  - Investigated temperature evolution and heat-transfer characteristics of a solid sphere enclosed by an air domain.
- **Finite Element Analysis Convergence and Mesh Independence** | Certificate

Internships

Solid Rocket Motor Thrust Vector Control (*Oct'22–2023*)
Defence Research and Development Organisation (DRDO)
Advanced Systems Laboratory, Hyderabad

- Studied solid rocket motor dynamics, thrust-vectoring principles and nozzle pressure variation.
- Designed an elliptical convergent–divergent nozzle in AutoCAD and subdivided the geometry into three sections.
- Studied thrust-vectoring mechanisms and their role in rocket trajectory control and stability.

Positions of Responsibility

- **Teaching Assistant, Engineering Graphics (Indian Institute of Technology, Kanpur)** (*Aug'2024–Jul'2025*)
  - Prepared written and computer-based assignments for undergraduate students.
  - Mentored 30 undergraduate students and reviewed their academic progress.
- **Volunteer, ACGT Conference** (*Aug'2024*)
  - Guided 80 participants during the conference.

Scholastic Achievements

- Secured **All India Rank 1017** in **GATE 2023**.
- Received the **Prathibha Puraskar Award** on SSC merit from the Government of Andhra Pradesh (2017).
- Received the **Ugam Legrand Scholarship** on intermediate merit basis (2019–2023).